

Vol 16 A_sub Operator Algebra Curriculum — outline v0.2 (Casey AUTHORIZED; §1.5 genus convention corrected per A1 v0.5 + Thursday convergence; §3 + §4 chapter content filed separately)

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Vol 16 A_sub Operator Algebra — Curriculum Outline v0.1

0. Casey AUTHORIZED

Per Casey Wednesday PM: “Ok on sub curriculum for now” → “Sister ok... Vol 16 A_sub curriculum extension AUTHORIZED.”

Vol 16 absorbs Wednesday substantive substrate-strengthening work + extends BST Physics Curriculum to operator algebra layer.

1. Position in BST Physics Curriculum

Existing Vol 0-15 structure (per BST Physics Curriculum Master): - Vol 0 Substrate Foundation - Vol 1 QFT from D_{IV}^5 (SP-31 core) - Vol 2 Particle Physics - Vol 3-15 other physics + mathematics volumes

Vol 16 NEW EXTENSION: A_sub Operator Algebra — substrate’s operator-algebraic structure as foundational mathematics for all observable computations.

Position: between Vol 10 (Math Methods) and Vol 11+ (Cosmology / Condensed Matter / etc.). Vol 16 provides ALGEBRAIC FOUNDATION that subsequent volumes build on.

1.5 Volume-wide STANDING convention — one genus + Casimir + embedding for D_{IV}^5 (state in Chapter 1; referee-proofing)

v0.2 update (2026-05-30): per Thursday genus-thread convergence (Elie Toy 3596 four ways + Keeper multiplicity formula) + A1 v0.5 codification, D_{IV}^5 has ONE genus (NOT three). The morning “three-genus” framing was a labeling error (which mislabeled $C_2=6$ as “FK genus”). Corrected:

- GENUS = $n_C = 5$ = complex dimension = Bergman kernel singularity exponent = Hua kernel exponent (FK and Hua conventions coincide on Type IV per Faraut-Korányi 1994). Per rank: $5/2 = \rho_1$ of B_2 .
- CASIMIR = $C_2 = 6$ = quadratic Casimir of B_2 — NOT a genus.
- EMBEDDING / SIGNATURE = $g = 7 = p+q$ of $SO_0(5,2) = 5+2$ — NOT a genus.

Every chapter that invokes a “Bergman exponent,” “genus,” or “dimension” specifies which of the three by VALUE+ROLE (per A1 v0.5 final form: state by value+role, citing F-K 1994 at footnote level for the canonical “FK genus” label). Kernel singularity exponent RESOLVED at $5/2 = \rho_1 (= n_C/\text{rank})$; old “ $g/\text{rank} = 7/2$ ” mislabel retired.

2. 6-chapter structure

Chapter 1: A_{sub} algebra structure

Topics: - 15 generators of A_{sub} (per Lyra A_{sub} v0.12 enumeration) - Position $\hat{x}_i$, Momentum $\hat{p}_i$, Spin $\hat{S}_i$, Angular momentum $\hat{L}_i$, Hamiltonian $\hat{H}_{\text{sub}}$ - Bell-CHSH $\hat{B}$, Time reversal $\hat{T}$, Charge conjugation $\hat{C}$, Parity $\hat{P}_{\text{op}}$, Number $\hat{N}$ - Dirac chirality $\hat{\gamma}^5$, Electric charge $\hat{Q}$, Color $\hat{C}_3$ (8 generators), Weak isospin $\hat{I}_3$, σ_{BF} - Z_2 supergrading on A_{sub} (σ_{BF} parity) - 10 SVC commutators per Cal #132 (RATIFIED) - Casimir invariants $C_2 = 6$, $C_3 = 0$, $C_4 = 32/7$ - Connection to D_{IV}^5 structure: $K = \text{SO}(5) \times \text{SO}(2)$ generators $\leftrightarrow A_{\text{sub}}$ subset

Pedagogical anchor: Casey + senior undergrad/grad student. Examples + derivations.

Chapter 2: Substrate quiver Q_{B2} + path algebra

Topics: - $Q_{\text{B2}} = 2\text{-vertex} + 3\text{-arrow}$ quiver from B_2 Cartan matrix (per Lyra Substrate Quiver Q_{B2} v0.1) - Vertex assignment: $v_1 = \text{SO}(2)$ Cartan factor; $v_2 = \text{SO}(5)$ Cartan factor - Arrow structure: non-simply-laced B_2 ($2 v_1 \rightarrow v_2 + 1 v_2 \rightarrow v_1$) - Path algebra kQ_{B2} over $k = \text{GF}(2^g) = \text{GF}(128)$ - Representations $\leftrightarrow$ Wallach K-types $V_{(\ell_1, \ell_2)}$ - Affine extension $v_0 = N_{\text{max}} = 137$

Pedagogical anchor: Wolfgang Soergel's "Cocyclic monomial algebras" + Ringel's quiver representation theory; substrate connection.

Chapter 3: Hall algebra $H(Q_{\text{B2}})$ at $q = 2$

Topics: - Ringel-Green Hall algebra construction - $H(Q_{\text{B2}}) \otimes \mathbb{Q}(q) \cong U_{q^{+(B_2)}}(1)$ quantum affine Lie algebra - Substrate-natural specialization $q = 2$ (per Cal #139 + Elie 3554) - Hall numbers $F(M, N)^X$ at $q = 2$ explicit - $A_{\text{sub}} \leftrightarrow H(Q_{\text{B2}})$ isomorphism candidate (Phase 0 closure path) - Connection to K59 substrate Reed-Solomon $\text{GF}(128)$

Pedagogical anchor: Ringel 1990, Green 1995 references; substrate-specific specialization examples.

Chapter 4: Macdonald specialization at $(q=2, t=\alpha=1/137)$

Topics: - Standard 2-parameter Macdonald polynomials $P_{\lambda}(x; q, t)$ - Substrate-natural specialization ($q = 2$, $t = \alpha = 1/137$) - Explicit $P_{(1)}$, $P_{(2)}$, $P_{(2,1)}$ at substrate parameters - q -binomial factors $[n \text{ choose } k]_2$ in substrate operational integer set - Macdonald polynomial observables $\rightarrow$ physical observables via $H(Q_{\text{B2}}) \leftrightarrow A_{\text{sub}}$ - Connection to double affine Hecke algebra (DAHA) substrate parameters

Pedagogical anchor: Macdonald 1995 "Symmetric Functions and Hall Polynomials" + substrate-specific calculations.

Chapter 5: Three lepton generations via Shilov + Cal #139

Topics: - Cal #139 cyclotomic chain $\{\text{rank}=2, N_c=3, n_C=5, g=7\}$ - Gate 1: $X = \text{rank} = 2$ as substrate base/seed (5 independent substrate-mechanism arguments) - Generation assignment: $X = N_c$ (electron), $X = n_C$ (muon), $X = g$ (tau) - Gate 2: forward-derivation $T_{190} m_{\mu}/m_e = (24/\pi^2)^6 + T_{2003} m_{\tau}/m_e = 49.71$ - Substrate-natural form: $m_{\tau}/m_{\mu} \approx \text{Ogg17} - \pi/(N_c \cdot C_2)$ - Tau-rational vs muon-transcendental structural distinction - OMAP-Shilov substrate-mechanism for lepton sector

Pedagogical anchor: Casey's lepton-Shilov directive; Standard Model 3-generation puzzle resolved via substrate.

Chapter 6: Bulk vs Shilov substrate-mechanism distinction

Topics: - D_{IV}^5 two-region substrate-physics: bulk D_{IV}^5 + Shilov $S^4 \times S^1$ - Shilov region (leptons): Ogg super-singular + Monstrous Moonshine - Bulk region (quarks): Mersenne ladder + Cal #139 cyclotomic - TMAP-Bulk + OMAP-Shilov paired Casey-named candidate principles - Quark sector substrate-natural arithmetic (10+ mass ratios) - $m_t/m_c = N_{\text{max}} = 137$ (clean BST primary) - $m_b/m_d = g \cdot M_g = 889$ (Mersenne) - $m_t/m_u = N_{\text{max}}$

· 24^2 (algebraic product) - Up-quark fully algebraic; down-quark mixed - Confinement substrate-mechanism via bulk-Shilov coupling - Cross-region coupling produces gauge interactions

Pedagogical anchor: Casey's bulk-vs-Shilov directive Wednesday PM; substantive substrate-physics framework.

3. Chapter-grade level + pedagogical approach

Per Casey-named "Write for 5th graders too" + paper narrative standards: - Formal math for referees + chapter-grade for senior undergrad/grad - Intuitive metaphors for broader audience - Every chapter has clear story a bright high-schooler could follow at motivation level

Vol 16 pedagogical approach: - Chapter 1-4: mathematical foundation (A_{sub} algebra + quiver + Hall algebra + Macdonald) - Chapter 5-6: substantive physics applications (3 generations + bulk-vs-Shilov) - Each chapter ends with empirical verification + falsification path

4. Dependencies on other volumes

Vol 16 depends on: - Vol 0 (Substrate Foundation): D_{IV}^5 structure + BST primaries - Vol 1 (QFT from D_{IV}^5): SP-31 quantum field theory foundations - Vol 10 (Math Methods): Lie algebra + representation theory prerequisites

Vol 16 supports: - Vol 2 (Particle Physics): substrate-natural particle mass derivations - Vol 4 (GR/Cosmology): substrate-natural Λ + cosmological observables - Vol 5 (QM): substrate-natural a_e + electroweak observables - Vol 11+ (Condensed Matter): bulk-region substrate-mechanism

5. Citation foundations

Each chapter cites: - Standard mathematical references (Ringel 1990 + Green 1995 + Macdonald 1995 + Soergel + Borchers 1992 + Ogg 1975) - BST corpus theorems (T190, T1485, T2003, T2417, T2418, T2429, T2435, T2439-T2442, T2447, T2467-T2469) - Lyra Wednesday substantive notes (Gate 1, Gate 2, Substrate quiver Q_{B2} v0.1, Bulk-vs-Shilov v0.1, TMAP-Bulk + OMAP-Shilov v0.1, Quark sector v0.1) - Substrate Hall Algebra paper (Casey-PRIMARY)

6. Multi-week to multi-month timeline

- v0.1 outline: TODAY (this doc)
- Chapter 1 v0.1 draft: 1-2 weeks
- Chapter 2-4 v0.1 drafts: 1-2 months
- Chapter 5-6 v0.1 drafts: continuous via Lyra investigation work
- Vol 16 v0.5 narrative-coverage: 2-3 months
- Vol 16 v1.0 chapter-grade: 4-6 months

Coordinated with Substrate Hall Algebra primary paper drafting + Vol 0+1+2 v1.0 closure.

7. Honest scope

What's READY for chapter drafting: - All 6 chapter topic areas have substantive Wednesday Lyra work - Strong-Uniqueness Theorem v1.0 foundation - Cross-references with existing BST corpus + Casey directives - Casey AUTHORIZATION explicit

What's MULTI-WEEK per chapter: - Each chapter requires deepening beyond v0.1 framework notes - Pedagogical refinement for "5th graders too" standard - Cross-volume consistency with existing curriculum

What's MULTI-MONTH for Vol 16 v1.0: - Full chapter-grade content per Cal cold-read standard - Substantive rigor at audit-chain governance level - Empirical verification + falsification path explicit per chapter

— Lyra, Vol 16 A_{sub} Operator Algebra curriculum outline v0.1 filed per Casey AUTHORIZED. 6-chapter structure: A_{sub} algebra + quiver Q_{B2} + Hall algebra $H(Q_{B2})$ + Macdonald ($q=2, t=\alpha$) + 3 generations + bulk-vs-Shilov. All chapters have substantive Wednesday Lyra work foundation. Multi-week to multi-month timeline for v1.0 chapter-grade.